

Energy Valve

Coil Estimator

A guided, five-step tool for a single coil. This manual walks every step, every field you can type in, and what each result means — including the financial figures, which are the ones most often misread.

Workbook	Belimo Energy Valve Coil Estimator_ML.xlsm
Languages	English · Español · Português · Français
Units	US or Metric
Scope	One coil, pumping savings only
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Belimo Americas · Internal field reference

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1 What the tool does

It estimates what you save on pumping when an Energy Valve holds a higher Delta-T across a single coil.

A coil is sized for a design temperature rise – its Delta-T. In practice most coils run at a lower Delta-T than they were selected for, which means the pump has to push **more water** through them to move the same amount of heat.

More flow costs money, and it costs it disproportionately: pump power varies with roughly the cube of flow, so cutting flow by a third cuts pump power by around 70%. An Energy Valve holds the coil at the Delta-T you ask for, so the flow — and the pumping bill — comes down.

This is a coil-level tool, and it counts pumping savings only. Its companion, the Energy Valve Savings Estimator, models a whole chilled-water plant and adds the chiller-side savings. Use this one when the conversation is about a specific coil or air handler; use the plant tool when it is about the central plant.

How the workbook is laid out

TAB	WHAT IT IS FOR
Home	Everything you fill in, in five numbered steps, plus the headline financial results.
Pump Savings	The calculation engine. One block per Delta-T, plus Energy Valve model selection.
Flow Curves	Reference curves relating control signal to flow.
Charts	The same relationships plotted.
Results	The customer-facing summary. This is the page you present.

2 Before you start

The step callouts

The Home tab is laid out as a wizard. Each orange section band has a numbered blue Step callout to its right telling you what to enter in that block. Work top to bottom, and the tool takes care of the rest.

Which cells can I type in?

SHADING	MEANING
Pale / near-white	Editable. Type here.
Blue-grey	Label or calculated result. Protected – the tool writes it.
Orange band	Section heading.

The sheets are protected so a stray keystroke cannot break a formula. If a cell will not accept typing, that is the tool telling you it is a result rather than an input.

Language and units

CONTROL	CELL	CHOICES AND EFFECT
Language	F5	English, Español, Português, Français. Translates every label, heading and unit word, on every tab.
Units	F6	US or Metric. Switches every flow, head, temperature, power and volume figure and its unit. The calculation always runs in US units and converts for display, so switching back and forth never drifts.

3 Home tab – the five steps

Everything you type lives here. Twenty-one fields, in five guided blocks.

Steps 1–2 · Language and coil design data

The screenshot shows the Belimo Energy Valve software interface. At the top, there's a header with the Belimo logo and the text "Belimo Energy Valve". Below this, there are two main sections: "Language & Unit Selection" and "Coil Design Input Data".

Language & Unit Selection: This section has two dropdown menus. The "Language" menu is set to "English" and the "Units" menu is set to "US".

Coil Design Input Data: This section contains a table with 21 fields. The fields are organized into two columns: "Field Name" and "Value/Unit".

Field Name	Value	Unit
Electricity Cost (Average)	0.10	\$/kWh
Design Coil Load	310	GPM
Number Of Coils	1	pcs.
Operating Hours / Day	20	hours
Operating Days / Week	6	days
Operating Weeks / Year	52	weeks
Actual Primary Pump Head (Design)	30	ft.
Pump, VFD, Motor Efficiency (Average)	70	%
Design Delta T	12	°F
Actual Delta T (Prior To Energy Valve Installation)	6	°F
Future Delta T (Energy Valve Delta T Manager Setting)	12	°F

On the right side of the interface, there are two steps indicated by blue buttons: "Step 1" and "Step 2".

Step 1: "Select your preferred Language".

Step 2: "Enter into the boxes on the left, the: Electric Costs, Coil Design GPM, Operating Hours, Pump Head, Motor Efficiency, DeltaT's".

Step 1 sets language and units. Step 2 is the coil itself – the values every later number is built from.

FIELD	CELL	UNIT	WHAT IT IS
Electricity Cost (Average)	F10	\$/kWh	Blended all-in rate. Every dollar figure in the workbook runs through this one number.
Design Coil Load	F11	GPM	The coil's design FLOW. Despite the word "Load", this is a flow rate in gallons per minute, not a tonnage – note the unit beside it.
Number Of Coils	F12	pcs.	Fixed at 1 – this tool estimates one coil at a time. Not editable.
Operating Hours / Day	F13	hours	With the two below, sets annual run hours.
Operating Days / Week	F14	days	
Operating Weeks / Year	F15	weeks	
Actual Primary Pump Head	F16	ft.	Head the pump works against at design flow.
Pump, VFD, Motor Efficiency	F17	%	Combined wire-to-water efficiency.
Design Delta T	F18	°F	What the coil was selected for.
Actual Delta T	F19	°F	What it achieves today, before the Energy Valve – the "before" case.
Future Delta T	F20	°F	What the Energy Valve's Delta-T Manager will hold – the "after" case.

The two Delta-T fields at the bottom are the whole estimate. Actual and Future are the before-and-after the entire comparison is built from, and the tool enforces a minimum gap between them so the pair always makes physical sense.

Steps 3–4 · Operating time and climate region

Operating Time Input Data

Operating Hours / Day	20	hours
Operating Days / Week	6	days
Operating Weeks / Year	52	weeks
Annual Operating Hours	5497	hours
Climate Region	North East	

Point	Bin	Load	% of Operating Time	Operating Hours
A	Design	100%	25.0%	1374
B	Peak	75%	50.0%	2748
C	Low	50%	25.0%	1374
D	Min	25%	0.0%	0
			100%	5497

Step 3

Enter into the boxes on the left, the: Climate Region from the drop down box

% of Operating Time
Enter Percentages of operating time.

Annual hours are computed from your three entries, then split across four load points.

Rather than simulate every hour of the year, the tool uses four representative operating points and gives each a share of the running time. Only the share column is editable, and the four must total 100%.

POINT	CELL	DEFAULT	MEANING
A – Design	E33	25%	Full-load hours.
B – Peak	E34	50%	The bulk of a typical year.
C – Low	E35	25%	Light load.
D – Minimum	E36	0%	Available if the coil genuinely idles for part of the year.

Climate Region Degree Day Input Data

Climate Region	Cooling Degree Days	Humidity Adjustment	Multiplier
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.93
South	2004	30%	1.71
South West	2750	5%	1.89
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.27
Custom Region	800	0%	0.52
Average US	1526	19%	1.19

Selected Value 0.88

Step 4

Verify your Climate Region or select custom and insert your own value.



U.S. Standard Regions

The region you pick supplies a multiplier; the Selected Value beneath the table is the one in play.

FIELD	CELL	NOTES
Climate Region	E30	Dropdown. Picks the cooling-degree-day profile for the site.
Custom Region	E53, F53	Editable. Enter your own cooling degree days and adjustment for a site that fits none of the standard regions, then select "Custom Region" above.

Step 5 · Financial inputs and results

Financial Inputs				
Investment Cost (Energy Valve + Installation)		\$4,000.00	\$	
Discount Rate		4.0%	%	
Estimated Measure Life		15	years	
		5 years	10 years	15 years
Net Present Value (NPV)	\$3,488	\$9,643	\$14,701	\$18,859
Savings-Investment Ratio (SIR)	1.87	3.41	4.68	5.71
Simple Payback Period	2.4 years			

Step 5

Enter the estimated cost to purchase Energy Valves and the cost of installation.

Three inputs, and the three results they drive, shown across four investment horizons.

FIELD	CELL	UNIT	WHAT IT IS
Investment Cost	F62	\$	Energy Valve plus installation. Use the INCREMENTAL cost over the valve that would otherwise have been fitted, not the whole valve package.
Discount Rate	F63	%	Converts future dollars into today's. Usually the organisation's cost of capital or required return.
Estimated Measure Life	F64	years	How long the measure is expected to keep saving. Feeds the lifetime figures on the Results tab.

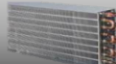
Below the inputs, three results appear at 5, 10, 15 and 20 years: the **Net Present Value (NPV)**, the **Savings-Investment Ratio (SIR)**, and the **Simple Payback Period**. All three are explained in section 7 – for now, treat Net Present Value as “what the project is worth in today's money” and the Savings-Investment Ratio as “how many dollars back per dollar in”.

4 Pump Savings tab

The engine. Nothing here needs filling in except one factor – but it is worth understanding what it is doing.

BELIMO

Belimo Energy Valve



Coil Design Data

Pump Design Factor

125%

Max Flow Rate

387.5 GPM

Editable Field

Label

Minimum Pump head

0

Design Chilled Water Coil Flow [V'max]

310

Design Chilled Water Coil ΔT

12

Design Chilled Water Coil [tons]

155

Design Chilled Water Coil [BTU/h]

1,860,000

Energy Valve Model

EV400SU-317

95..317 gpm

4" DN100 * based on Vmax

Coil Design Performance Data

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft.]	Power [HP]	BTU/h	Energy [kWh]	Annual gallons
100	25.0	1374	310	30	3.4	1,860,000	3,439	25,559,179
75	50.0	2748	233	17	1.4	1,395,000	2,902	38,338,769
50	25.0	1374	155	8	0.4	930,000	430	12,779,590
25	0.0	0	78	2	0.1	465,000	0	0

6,771

76,677,537

677

Pump Costs

\$344

Pump costs

\$677

Annual gallons

76,677,537

Avg GPM

233

Energy [kWh]

6,771

Pump Affinity Laws

$$\frac{RPM_1}{RPM_2} = \left(\frac{GPM_1}{GPM_2}\right)^3 = \left(\frac{HEAD_1}{HEAD_2}\right)^{1.5} = \left(\frac{HP_1}{HP_2}\right)^{0.5}$$

HP = GPM x HD x Specific Gravity
3960 x Pump Efficiency

Chilled Water Coil Flows at different Delta T [°F]

Chilled Water Coil Flow at Delta T [°F]

4

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft.]	Power [HP]	BTU/h	Energy [kWh]	Annual gallons
100	25.0	1374	388	47	6.6	775,000	6,717	31,948,974
75	50.0	2748	388	47	6.6	775,000	13,435	63,897,948
50	25.0	1374	388	47	6.6	775,000	6,717	31,948,974
25	0.0	0	233	17	1.4	465,000	0	0

26,869

127,795,895

Effects of Low ΔT vs Design

BTU/h As % of Design	Increase In GPM	Additional Pumped gallons	Additional Pump Costs	Pump costs gallons	Avg GPM	Energy [kWh]	Total Annual Pump costs
42%	78	6,389,795	\$672				\$2,687
56%	155	25,559,179	\$1,343				127,795,895
83%	233	19,169,384	\$672				388
100%	155	0	\$0				26,869

2,687

Design data at the top, then a calculation block for each Delta-T in the comparison range.

FIELD	CELL	WHAT IT IS
Pump Design Factor	F5	The only editable cell on this tab (default 1.25). A safety margin on design flow – how much bigger the pump is than the coil strictly needs. Leave it alone unless you know the real oversizing.

How a block reads

Chilled Water Coil Flows at different Delta T [°F]

Chilled Water Coil Flow at Delta T [°F]

4

Load [%]	Time [%]	Time [h]	Flow [GPM]	Head [ft.]	Power [HP]	BTU/h	Energy [kWh]	Annual gallons
100	25.0	1374	388	47	6.6	775,000	6,717	31,948,974
75	50.0	2748	388	47	6.6	775,000	13,435	63,897,948
50	25.0	1374	388	47	6.6	775,000	6,717	31,948,974
25	0.0	0	233	17	1.4	465,000	0	0

26,869

127,795,895

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100%	155	0	\$0				26,869

2,687

Each block is headed by its Delta-T and carries the four load points across flow, head, power, energy and cost.

There is one block per Delta-T in the range, and they all work the same way: for each of the four load points the tool works out the flow needed at that Delta-T, the head the pump sees at that flow, the resulting pump power, and from the hours in that bin, the energy and the cost. Add the four rows and you have that Delta-T's annual pumping bill.

Energy Valve selection

1.65	EV050	1.65..6.6 gpm	0.5" DN15	6.2..25 lpm
6.6	EV075	2.7..11 gpm	0.75" DN20	10.2..41.6 lpm
12	EV100	4.5..18.2 gpm	1" DN25	17..68.9 lpm
19	EV125	7.1..28.5 gpm	1.25" DN32	26.9..107.9 lpm
29	EV150	11..44 gpm	1.5" DN40	41.6..166.6 lpm
45	EV200	16.5..66 gpm	2" DN50	62.5..249.8 lpm
67	EV200H	25..100 gpm	2" DN50	94.6..378.5 lpm
101	EV250SU-127	38..127 gpm	2.5" DN65	143.8..480.7 lpm
128	EV300SU-180	54..180 gpm	3" DN80	204.4..681.4 lpm
181	EV400SU-317	95..317 gpm	4" DN100	359.6..1200 lpm
318	EV500SU-495	149..495 gpm	5" DN125	564..1873.8 lpm
496	EV600SU-713	214..713 gpm	6" DN150	810.1..2699 lpm
714	Dual EV's	Up to 1400 gpm		Up to 5299.6 lpm

The model table. The tool matches your design flow to a valve and shows its capacity range.

Using the design flow you entered in Step 2, the tool picks the Energy Valve model that covers it and displays that model with its flow capacity range. The range is shown in whichever unit system you have selected.

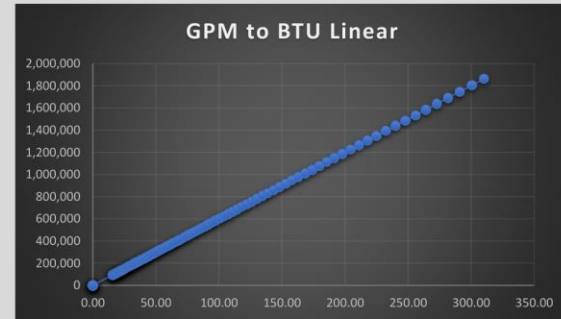
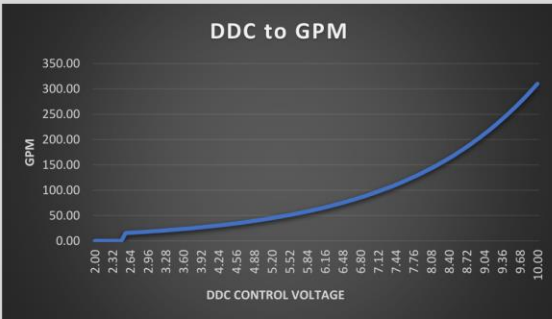
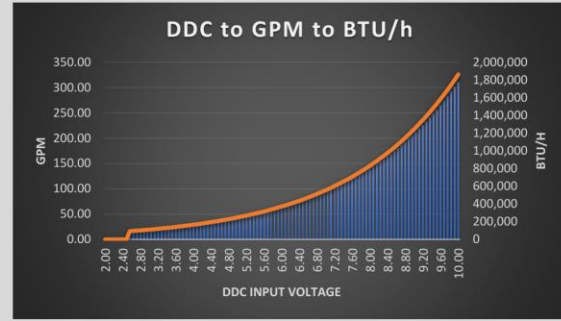
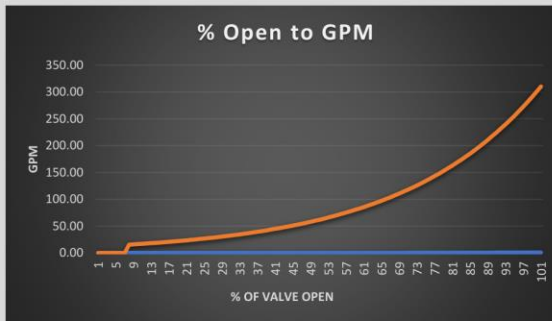
5 Flow Curves and Charts

Reference material. Nothing to fill in on either tab.

 Belimo Energy Valve								
Used to Calculate Projected Flow								
DDC Control signal			Valve Position & Flow based on DDC Voltage					
	Voltage	Voltage	Linear Part	Equal Part	V'max	V'max	ΔT	btuH
	0.50	2.00			Linear	EQ %		
	10.00	10.00			310	310	12	
In %	In V	In V	In %	In %	g/min.	g/min.	°F	btu/H
0	0.50	2.00	0.0%	4.0%	0.00	0.00		0
1	0.60	2.08	0.4%	4.1%	0.00	0.00		0
2	0.69	2.16	0.7%	4.3%	0.00	0.00		0
3	0.79	2.24	1.1%	4.4%	0.00	0.00		0
4	0.88	2.32	1.4%	4.5%	0.00	0.00		0
5	0.98	2.40	1.8%	4.7%	0.00	0.00		0
6	1.07	2.48	2.1%	4.9%	0.00	0.00		0
7	1.17	2.56	2.5%	5.0%	0.00	15.54		93,210
8	1.26	2.64	2.8%	5.2%	8.68	16.04		96,260
9	1.36	2.72	3.2%	5.3%	9.77	16.57		99,410
10	1.45	2.80	3.5%	5.5%	10.85	17.11		102,663
11	1.55	2.88	3.9%	5.7%	11.94	17.67		106,023
12	1.64	2.96	4.2%	5.9%	13.02	18.25		109,492
13	1.74	3.04	4.6%	6.1%	14.11	18.85		113,076
14	1.83	3.12	4.9%	6.3%	15.19	19.46		116,776
15	1.93	3.20	5.3%	6.5%	16.28	20.10		120,597
16	2.02	3.28	5.6%	6.7%	17.36	20.76		124,544

Flow Curves relates the control signal to actual flow through the valve.

These tabs are there for the technical conversation – how a control signal maps to valve position and flow, and what that looks like plotted. They are useful when a customer or controls contractor asks how the valve behaves across its range; they are not part of the savings calculation.



The same relationships, plotted.

6 Results tab

The customer-facing page. Nothing to fill in – this is what you present or print.

Energy Valve Savings

Before & After: Changing Your Delta-T

	Actual (Before)	Future (After Energy Valve)
Delta-T [°F]	6	12
Avg Flow [GPM]	368	233
Pumping Cost / Year	\$2,359	\$677
Energy / Year [kWh]	23,591	6,771

Delta-T [°F]	Avg [GPM]	Avg [tons]	Pumping Cost	Energy [kWh]	Annual gallons
4	388	65	\$2,687	26,869	127,795,895
6	368	92	\$2,359	23,591	121,406,100
8	329	110	\$1,796	17,962	108,626,511
10	279	116	\$1,170	11,700	92,013,045
12	233	116	\$677	6,771	76,677,537
14	199	116	\$426	4,264	65,723,603
Annual Energy Savings for changing Delta T from 6. to 12.			\$1,682	16,820	44,728,563
Calculate Annual Savings by Changing the DeltaT using the slider					
From	To				
6	14		\$1,933	19,327	55,682,497

Before and after, the full Delta-T range colour-graded from worst to best, and the interactive slider beneath.

What each block shows

BLOCK	CONTENTS
Before & After	Your Actual and Future Delta-T side by side: Delta-T, average flow, pumping cost per year and energy per year.
Delta-T table	Six scenarios across the range, colour-graded red (worst) to blue (best), with average flow, average tonnage, pumping cost, energy and annual water volume.
Annual Energy Savings	The headline line: dollars, kilowatt-hours and gallons saved by moving from your Actual Delta-T to your Future one.
The slider	Set any "To" Delta-T and the savings line beneath recalculates live. The "From" case stays pinned to the coil's actual Delta-T.

The slider is the most persuasive thing on the tab: it answers “what if we only get halfway there?” while the customer watches, instead of forcing a single optimistic number.

Financial and environmental summary

Greenhouse Gas Equivalents

Annual Carbon Dioxide Avoided	26,155	lbs
Number of Tree Seedlings Grown for 10 Years	304	tree seedlings
Equivalent to Number of Houses' Annual Electricity Use	1.6	homes

Financial Summary

	5 years	10 years	15 years	20 years
Net Present Value (NPV)	\$3,488	\$9,643	\$14,701	\$18,859
Savings-Investment Ratio (SIR)	1.87	3.41	4.68	5.71
Simple Payback Period	2.4 years			

The same three financial results as the Home tab, plus the emissions avoided.

The greenhouse-gas block expresses the saving as pounds of carbon dioxide avoided and as the two equivalents people can actually picture – tree seedlings grown for ten years, and a number of homes' annual electricity use.

Utility rebate support

Utility Rebate Support

Measure & Operating Basis

Baseline Delta-T (before)	6.0	°F
Post-Retrofit Delta-T (after)	12.0	°F
Number of Coils	1	
Annual Operating Hours	5,497	hours

Energy & Demand Savings

Annual Electricity Savings	16,820	kWh
Average Demand Reduction	3.060	kW
Demand Reduction at Peak Load	2.385	kW
Annual Cost Savings	1,682	\$
Baseline Annual Pump Energy	23,591	kWh
Post-Retrofit Annual Pump Energy	6,771	kWh
Pump Energy Reduction	71.3%	

Lifetime & Environmental

Estimated Measure Life	15	years
Lifetime Electricity Savings	252,301	kWh
Annual Water Volume Reduction	44,728,563	gallons
Annual Carbon Dioxide Avoided	26,155	lbs

Savings are calculated from pump affinity laws applied to the coil's load profile across the operating year. Baseline and post-retrofit pump energy are computed at the Actual and Future Delta-T respectively, using the site's own operating hours, pump head,

Laid out in the form incentive programmes ask for, with a methodology note at the foot.

This block restates the result in the shape a utility programme wants: the measure basis (baseline and post-retrofit Delta-T, coil count, annual hours), the energy and demand savings, and the lifetime and

environmental figures. It closes with a note describing the calculation basis, so a reviewer can see the method without having to ask.

Two demand figures, and they are not the same thing. Average Demand Reduction is the annual energy saving divided by annual hours. Demand Reduction at Peak Load is the difference in pump power at the design point. A rebate form usually wants the peak figure; do not quote one where the other is asked for.

7 The financial results

Three numbers, shown at four horizons. These are the ones that get misread, so they are worth a careful read.

FIRST, THE IDEA BEHIND ALL THREE

Present value

Money in the future is worth less than money now — you could have invested today's dollar in the meantime. “Present value” is what a future amount is worth today, once you have discounted it back at whatever rate you set in Step 5.

Why it matters here: the discount rate is a lever. Raise it and future savings count for less, so both of the first two results below get smaller. It is not a fudge factor — set it to what your organisation actually requires and leave it there.

RESULT 1

Net Present Value (NPV)

Take every year's saving, discount it back to today, add them up, and subtract what you spent on day one. What is left is the Net Present Value: the value the project adds, expressed in today's dollars.

Above zero → worth doing. Below zero → not at this discount rate.

Below zero does not mean the project loses money. It means it does not clear the return you demanded of it. Lower the required return and the same project can turn positive.

On the sheet: shown at 5, 10, 15 and 20 years. It grows with the horizon, because the spend happens once at the start while the savings keep accruing.

RESULT 2

Savings-Investment Ratio (SIR)

The present value of everything you get back, divided by what you put in. Because it is a ratio rather than an amount, it lets you compare projects of very different sizes on equal terms.

Above 1.0 → returns more than it costs. Below 1.0 → does not pay back in present-value terms.

A Savings-Investment Ratio of 1.9 means every dollar invested returns about \$1.90 in today's money. Many public-sector and utility incentive programmes set a minimum ratio as an eligibility condition, which is why it sits alongside Net Present Value on both the Home and Results tabs.

RESULT 3

Simple Payback Period

How long the savings take to repay the investment, ignoring the time value of money entirely. The number a building owner asks for first, and the easiest to explain.

Read it as a sanity check, not as the whole case. It ignores everything that happens after payback, so it systematically undersells a long-lived measure. Quote it because people ask for it — then show the Net Present Value.

Which one do you lead with? For a building owner, Simple Payback then Net Present Value. For a finance department, Net Present Value. For a utility or public-sector programme, the Savings-Investment Ratio — it is often written straight into the eligibility rules.

Reading the four horizons

Each result is repeated at 5, 10, 15 and 20 years, answering the same question over a different holding period: if we judged this on an N-year view, how would it look? Short horizons always look worse, because the money goes out on day one while the savings accumulate. Quote the horizon that matches how the customer actually evaluates capital — and say which horizon you are quoting.

8 Practicalities

Presenting and printing

The Results tab is the page to show. Print or export it once the inputs are settled — it carries the comparison, the savings, the financial summary and the rebate block on one sheet.

Language

Switching language on the Home tab re-labels the whole workbook, including the Results tab headings and the chart legend. Numbers and units follow the separate Units dropdown.

If something looks wrong

SYMPTOM	CHECK
A cell will not let you type	It is a calculated result. Compare it against the shading key in section 2.
Savings look implausibly large	Check the gap between Actual and Future Delta-T, and the Pump Design Factor on the Pump Savings tab.
The financial results look wrong	Almost always the investment figure. Confirm it holds the incremental cost of the Energy Valve over the valve that would have been fitted anyway.
Delta-T entries are rejected	The tool enforces a minimum gap between Actual and Future, and a sensible range on Design. Adjust the pair together.
Numbers did not change after an edit	Click away from the cell to commit the entry, or press F9 to recalculate.

This tool produces an estimate. Results depend on the accuracy of the inputs and on conditions specific to the installation. Questions, corrections and feature requests go to Rick.Smith@us.belimo.com.